RACE
Regular Analysis Through Continuous Exercise

Hyperbola : RACE-01

Single Correct Type Questions:

1. The eccentricity of the hyperbola $4x^2 - 9y^2 - 8x = 32$ is -

(A) $\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{13}}{2}$ (D) $\frac{3}{2}$

MHB200

2. If foci of a hyperbola are foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. If the eccentricity of the hyperbola be 2, then its equation is -

(A) $\frac{x^2}{4}-\frac{y^2}{12}=1$ (B) $\frac{x^2}{12}-\frac{y^2}{4}=1$ (C) $\frac{x^2}{12}+\frac{y^2}{4}=1$ (D) none of these

MHB201

Subjective Type Questions:

3. Find the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ which passes through $(4, 0)$ & $(3\sqrt{2}, 2)$

MHB202

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4. Find the equation to the hyperbola, whose eccentricity is $\frac{5}{4}$, focus is $(a, 0)$ and whose directrix is $4x - 3y = a$.

MHB203

5. In the hyperbola $4x^2 - 9y^2 = 36$, find eccentricity, co-ordinates of the foci, length of the axes and the latus rectum.

MHB204

6. Find the equation to the hyperbola, the distance between whose foci is 16 and whose eccentricity is $\sqrt{2}$. Also Transverse and conjugate axes coincides with the coordinate axes.

MHB205

7. Consider the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$. Find the following:

(a) centre **(b)** eccentricity
(c) focii **(d)** equation of directrix
(e) length of the latus rectum

MHB206

8. Find the equation to the hyperbola, referred to its axes as axes of coordinates,
(a) whose transverse and conjugate axes are respectively 3 and 4,
(b) whose conjugate axis is 5 and the distance between whose foci is 13
(c) whose conjugate axis is 7 and which passes through the point (3, –2)

MHB207

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9. Find the equation to the hyperbola of given transverse axis whose vertex bisects the distance between the centre and the focus.

MHB208

10. Find eccentricity of conjugate hyperbola of hyperbola $4x^2 - 16y^2 = 64$, also find area of quadrilateral formed by foci of hyperbola & its conjugate hyperbola

MHB209

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ANSWER KEY

1. B **2.** A **3.** $\sqrt{3}$

4. $(7y^2 + 24xy - 24ax - 6ay + 15a^2 = 0)$ **5.** $(6, 4; (\pm\sqrt{13}, 0); \sqrt{13}/3; 8/3)$

6. $(x^2 - y^2 = 32)$

7. (a) (–4,–1) ; (b) $\frac{5}{4}$; (c) (1,–1), (–9,–1); (d) $5x + 4 = 0, 5x + 36 = 0$; (e) $\frac{9}{2}$

8. (a) $16x^2 - 9y^2 = 36$ (b) $25x^2 - 144y^2 = 900$ (c) $65x^2 - 36y^2 = 441$

9. $(3x^2 - y^2 = 3a^2)$ **10.** ($\sqrt{5}$ & 40 sq. units)

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Hyperbola : RACE-02

Single Correct Type Questions:

1. If the line $y = 5x + 1$ touch the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ $\{b > 4\}$, then -

(A) $b^2 = \frac{1}{5}$ (B) $b^2 = 99$

(C) $b^2 = 4$ (D) $b^2 = 100$

MHB210

Subjective Type Questions:

2. Let e_1 be the eccentricity of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$, which passes through the foci of the hyperbola. If $e_1 e_2 = 1$, then find the length of the chord of the ellipse parallel to the x-axis and passing through (0,2).

MHB211

3. Let the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{\ell}{m}\left(1 + \sqrt{n}\right)$, where ℓ and m are co-prime numbers, then $\ell^2 + m^2 + n^2$ is equal to

MHB212

4. Show that the line $x\cos\alpha + y\sin\alpha = p$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$.

MHB213

5. Find the points common to the hyperbola $25x^2 - 9y^2 = 225$ and the straight line $25x + 12y - 45 = 0$.

MHB214

6. For what value of λ, does the line $y = 3x + \lambda$ touch the hyperbola $9x^2 - 5y^2 = 45$?

MHB215

7. Find the equation of the tangent to the hyperbola $4x^2 - 9y^2 = 1$, which is parallel to the line $4y = 5x + 7$.

MHB216

8. Find the equation of the tangent to the hyperbola $16x^2 - 9y^2 = 144$ at $\left(5, \frac{16}{3}\right)$.

MHB217

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9. Find the common tangent to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ and an ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$.

MHB218

10. If m_1 and m_2 are the slopes of the tangents to the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ which pass through the point (6, 2), then find the value of $11m_1m_2$ and $11(m_1 + m_2)$.

MHB219

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ANSWER KEY

1.	B	**2.**	$\frac{10\sqrt{5}}{3}$	**3.**	182	**5.**	$5, -\frac{20}{3}$
6.	$\lambda = \pm 6$	**7.**	$24y = 30x \pm \sqrt{161}$	**8.**	$5x - 3y = 9$	**9.**	$y = \pm x \pm \sqrt{7}$
10.	20 & 24						

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Hyperbola : RACE-03

Single Correct Type Questions:

1. For $0 < \theta < \pi/2$, if the eccentricity of the hyperbola $x^2 - y^2\operatorname{cosec}^2\theta = 5$ is $\sqrt{7}$ times eccentricity of the ellipse $x^2\operatorname{cosec}^2\theta + y^2 = 5$, then the value of θ is:

MHB220

2. Find the equation of normal to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ at (5, 0).

MHB221

3. Find the equation of normal to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at the point $\left(6, \frac{3}{2}\sqrt{5}\right)$.

MHB222

4. Find the condition for the line $\ell x + my + n = 0$ to be normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

MHB223

5. Find the equation of normal to the hyperbola $3x^2 - 4y^2 = 12$ having slope 1.

MHB224

6. If a chord joining the points $P(a\sec\theta, a\tan\theta)$ & $Q(a\sec\phi, a\tan\phi)$ on the hyperbola $x^2 - y^2 = a^2$ is a normal to it at P, then show that $\tan\phi = \tan\theta\,(4\sec^2\theta - 1)$.

MHB225

7. Find the equation to the chords of the hyperbola $x^2 - y^2 = 9$ which is bisected at (5, –3)

MHB226

8. Find the locus of the mid points of the chords of the circle $x^2 + y^2 = 16$ which are tangents to the hyperbola $9x^2 - 16y^2 = 144$.

MHB227

9. The asymptotes of a hyperbola are parallel to lines $2x + 3y = 0$ and $3x + 2y = 0$. The hyperbola has its centre at (1, 2) and it passes through (5, 3). Find its equation.

MHB228

10. If 'A' denotes pair of asymptotes of hyperbola 'H'. Find equation of conjugate hyperbola of H when $x^2 + 2xy - x - 2y = 0$ represents A and 'H' passes through (–1,1).

MHB229

ANSWER KEY

1.	$\frac{\pi}{3}$	**2.**	$y = 0$	**3.**	$8\sqrt{5}x + 18y = 75\sqrt{5}$
4.	$\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{(a^2+b^2)^2}{n^2}$	**5.**	$y = x \pm 7$	**7.**	$5x + 3y = 16$
8.	$(x^2 + y^2)^2 = 16x^2 - 9y^2$	**9.**	$(2x + 3y - 8)(3x + 2y - 7) = 154$		
10.	$(x^2 + 2xy - x - y - 2 = 0)$				

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Hyperbola : RACE-04

Multiple Type Questions:

1. If two tangents can be drawn to the different branches of hyperbola $x^2 - \frac{y^2}{4} = 1$ from the (α, α^2), then $\alpha \in$

(A) $(-\infty, -3)$ (B) $(3, \infty)$ (C) $(-2,0) \cup (0,2)$ (D) $(2, \infty)$

MHB230

Subjective Type Questions:

2. If equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola then write required conditions.

MHB231

3. Find the equation of hyperbola having foci $S_1(2, 1)$ and $S_2(10, 1)$ and a straight line $x + y - 9 = 0$ as its tangent. Also, find the equation of its director circle.

MHB232

4. A ray emanating from the point (5, 0) is incident on the hyperbola $9x^2 - 16y^2 = 144$ at the point $P(8, 3\sqrt{3})$. Find the equation of the reflected ray after first reflection.

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MHB233

5. Find the equation of tangent at the point (1, 2) to the rectangular hyperbola $xy = 2$.

MHB234

6. Prove that the locus of point, tangents from where to hyperbola $x^2 - y^2 = a^2$ inclined at an angle α & β with x-axis such that $\tan\alpha \tan\beta = 2$ is also a hyperbola. Find the eccentricity of this hyperbola.

MHB235

7. Find everything for the conic $xy = -36$.

MHB236

8. Consider hyperbola $xy = 16$ to find the following:

(i) Equation of tangent at point (2, 8)

(ii) Equation of chord of contact w.r.t. point (2, 3)

MHB237

(iii) Equation of chord which gets bisected at point (5, 6)

(iv) Equation of normal having slope 2.

MHB238

9. Consider the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$. Find the following:

(a) equation of auxiliary circle **(b)** equation of director circle

MHB239

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10. The portion of tangent intercepted between tangent at vertices of hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ subtends angle θ at focus, find all possible values of θ.

MHB240

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ANSWER KEY

1. A,B,D

2. $\Delta \neq 0, h^2 > ab, a + b = 0$

3. $\frac{(x-6)^2}{10}-\frac{(y-1)^2}{6}=1$ and $(x-6)^2 + (y-1)^2 = 4$

4. $3\sqrt{3}x-13y+15\sqrt{3}=0$

5. $2x + y = 4$

6. $e=\sqrt{3}$

7. Eccentricity $=\sqrt{2}$,
Asymptotes: $x = 0$ & $y = 0$,
Transverse axis $y = -x$, Conjugate axis: $y = x$,
Centre (0,0),
Vertices (6, –6) & (–6, 6),
Foci $\left(6\sqrt{2},-6\sqrt{2}\right)$ & $\left(-6\sqrt{2},6\sqrt{2}\right)$,
Length of LR $=12\sqrt{2}$,
Equation of auxiliary circle $x^2 + y^2 = 72$,
Equation of director circle $x^2 + y^2 = 0$,
Equation of directrices $x - y = \pm 6\sqrt{2}$.

8. (i) $4x + y - 16 = 0$ (ii) $3x + 2y - 32 = 0$
(iii) $6x + 5y = 60$ (iv) $2x-y=3\sqrt{8}$ and $2x-y=-3\sqrt{8}$

9. (a) $(x + 4)^2 + (y + 1)^2 = 16$; (b) $(x + 4)^2 + (y + 1)^2 = 7$

10. $\frac{\pi}{2}$

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